AI-Powered Enterprise E-Commerce Platform

Project Management Plan

Document Information

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Revision History

Version	Date	Description
1.0	DD/MM/YYYY	Initial Version

Table of Contents

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1. Introduction

This document defines the overall project management strategy for developing the AI-Powered Enterprise E-Commerce Platform.

The project combines:

- Enterprise Software Engineering
- Artificial Intelligence
- Microservices Architecture
- Cloud-Native Development Practices

The purpose of this document is to ensure the project is completed successfully within the available time and resources.

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2. Purpose

The objectives of this project plan are:

- Define project activities.
- Establish milestones.
- Identify resources.
- Manage risks.
- Monitor progress.
- Ensure successful delivery.

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3. Project Vision

Develop an enterprise-grade intelligent e-commerce platform capable of:

- supporting online shopping,
- providing AI-powered recommendations,
- detecting fraudulent activities,
- forecasting future sales,
- generating business analytics.

The system should demonstrate modern industry practices and portfolio-level engineering standards.

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4. Project Objectives

Business Objectives

OBJ-01

Improve customer experience.

OBJ-02

Increase sales using recommendations.

OBJ-03

Reduce fraudulent transactions.

OBJ-04

Support business intelligence.

Technical Objectives

OBJ-05

Implement microservices architecture.

OBJ-06

Integrate Java and Python services.

OBJ-07

Deploy containerized services.

OBJ-08

Implement enterprise security.

Educational Objectives

OBJ-09

Demonstrate full software engineering lifecycle.

OBJ-10

Build an industry-level portfolio project.

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5. Project Scope

In Scope

Functional Features

- Authentication
- Product Management
- Inventory Management
- Cart
- Wishlist
- Orders
- Payments
- Notifications

AI Features

- Recommendation System
- Fraud Detection
- Sales Forecasting
- Customer Segmentation
- AI Assistant

Engineering Features

- Docker Deployment
- Monitoring
- Logging
- CI/CD
- Security Architecture

Out of Scope (Initial Version)

- Native Mobile Applications
- Multi-vendor Marketplace
- Blockchain Payments
- Multi-region Deployment
- International Tax Systems

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6. Development Methodology

The project will follow:

Agile Scrum Methodology

Reasons:

- Incremental development
- Easier requirement management
- Continuous feedback
- Better risk control

Sprint Duration

2 Weeks

Scrum Events

Sprint Planning

Daily Standups

Sprint Review

Sprint Retrospective

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7. Work Breakdown Structure (WBS)

```
1. Requirements Engineering
2. Architecture Design
3. Database Design
4. API Design
5. Backend Development
6. AI Development
7. Frontend Development
8. Integration
9. Testing
10. Deployment
11. Documentation
```

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8. Project Phases

Phase 1 – Requirements Engineering

Activities:

- Vision
- SRS
- Feasibility Study
- Risk Analysis

Deliverables:

- Requirements Documents

Duration:

2 Weeks

Phase 2 – System Design

Activities:

- Architecture Design
- Database Design
- API Design

Deliverables:

- Design Documents

Duration:

2 Weeks

Phase 3 – Environment Setup

Activities:

- GitHub Repository
- Docker Setup
- Project Structure
- CI/CD Preparation

Duration:

1 Week

Phase 4 – Backend Development

Activities:

- Authentication Service
- Product Service
- Order Service
- Inventory Service

Duration:

4 Weeks

Phase 5 – AI Development

Activities:

- Dataset Collection
- Data Engineering
- Model Training
- Recommendation System
- Fraud Detection
- Forecasting

Duration:

4 Weeks

Phase 6 – Frontend Development

Activities:

- UI Design
- React Components
- API Integration

Duration:

3 Weeks

Phase 7 – Integration Phase

Activities:

- Java ↔ Python Integration
- RabbitMQ Integration
- End-to-End Testing

Duration:

2 Weeks

Phase 8 – Testing Phase

Activities:

- Unit Testing
- Integration Testing
- Security Testing
- Performance Testing

Duration:

2 Weeks

Phase 9 – Deployment Phase

Activities:

- Docker Deployment
- Monitoring Setup
- Production Configuration

Duration:

1 Week

Phase 10 – Final Documentation

Activities:

- User Documentation
- Technical Documentation
- Project Report

Duration:

1 Week

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9. Deliverables

Documentation Deliverables

- Vision Document
- SRS
- Architecture Documents
- Database Design
- API Specifications

Software Deliverables

- Frontend Application
- Backend Services
- AI Services
- Database Scripts
- Docker Configurations

AI Deliverables

- Trained Models
- Evaluation Reports
- Datasets
- Model Documentation

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10. Major Milestones

Milestone	Target
Requirements Completed	Week 2
Architecture Completed	Week 4
Backend Completed	Week 9
AI Models Completed	Week 11
Frontend Completed	Week 12
Integration Completed	Week 14
Testing Completed	Week 16
Deployment Completed	Week 17
Final Documentation Completed	Week 18

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11. Proposed Timeline

```
Weeks 1-2   → Requirements
Weeks 3-4   → Design
Week 5      → Environment Setup
Weeks 6-9   → Backend
Weeks 8-11  → AI Development
Weeks 10-12 → Frontend
Weeks 13-14 → Integration
Weeks 15-16 → Testing
Week 17     → Deployment
Week 18     → Final Documentation
```

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12. Team Roles and Responsibilities

Project Manager

Responsibilities:

- Planning
- Scheduling
- Monitoring

Backend Developer

Responsibilities:

- Spring Boot Services
- APIs
- Security

AI Engineer

Responsibilities:

- Dataset Preparation
- Model Development
- AI APIs

Frontend Developer

Responsibilities:

- React Development
- UI Integration

DevOps Engineer

Responsibilities:

- Docker
- CI/CD
- Monitoring

QA Engineer

Responsibilities:

- Testing
- Quality Assurance

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13. Resource Planning

Hardware Resources

- Development Laptop
- Internet Connectivity

Software Resources

- VS Code
- IntelliJ IDEA
- Docker
- PostgreSQL
- GitHub

Cloud Resources

Future:

- AWS
- Azure
- Kubernetes

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14. Risk Management Plan

Major risks:

- Requirement changes

- Dataset quality issues
- Integration complexity
- Security vulnerabilities
- Schedule delays

Mitigation:

- Incremental development
- Early testing
- Continuous documentation
- Regular reviews

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15. Communication Plan

Activity	Frequency
Progress Review	Weekly
Sprint Review	Every 2 Weeks
Risk Review	Weekly
Documentation Review	Weekly

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16. Quality Management Plan

Quality practices:

- Coding standards
- Code reviews
- Automated testing
- Static analysis
- Documentation reviews

Quality Metrics

Examples:

- Code coverage > 80%
- API response time < 2 seconds
- Recommendation accuracy targets

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17. Configuration Management

Tools:

- Git
- GitHub

Practices:

- Feature branches
- Pull requests
- Version tagging
- Release management

Branching Strategy

```
main
develop
feature/*
hotfix/*
release/*
```

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18. Change Management

Changes shall follow:

1. Change Request
2. Impact Analysis
3. Approval
4. Implementation
5. Verification

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19. Project Monitoring Metrics

Examples:

Schedule Metrics

- Tasks completed
- Sprint velocity

Quality Metrics

- Defect counts
- Test coverage

AI Metrics

- Recommendation accuracy
- Fraud detection accuracy
- Forecasting error

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20. Success Criteria

The project will be considered successful if:

Functional Success

All critical features are implemented.

Technical Success

Architecture functions correctly.

AI Success

Models achieve acceptable performance.

Educational Success

Demonstrates enterprise engineering capabilities.

Portfolio Success

Suitable for GitHub and internship applications.

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21. Assumptions

- Required technologies remain available.
- Public datasets can be collected.
- Development environment remains stable.
- Team members continue learning.

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22. Constraints

- Limited time.
- Limited hardware resources.
- Learning curve.
- Dataset availability.

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23. Future Roadmap

Possible future phases:

Phase 2

- Mobile Application
- Kafka Integration
- Kubernetes Deployment

Phase 3

- Multi-vendor Marketplace
- Event-Driven Architecture
- Generative AI Assistant

Phase 4

- Real-Time Recommendation System
- Data Warehouse
- Big Data Analytics

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24. Conclusion

This project plan provides a structured roadmap for developing the AI-Powered Enterprise E-Commerce Platform.

The phased approach ensures:

- manageable complexity,
- reduced risks,
- continuous progress,
- and high-quality deliverables.

The project is expected to provide substantial educational, technical, and portfolio value while demonstrating enterprise software engineering practices.

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